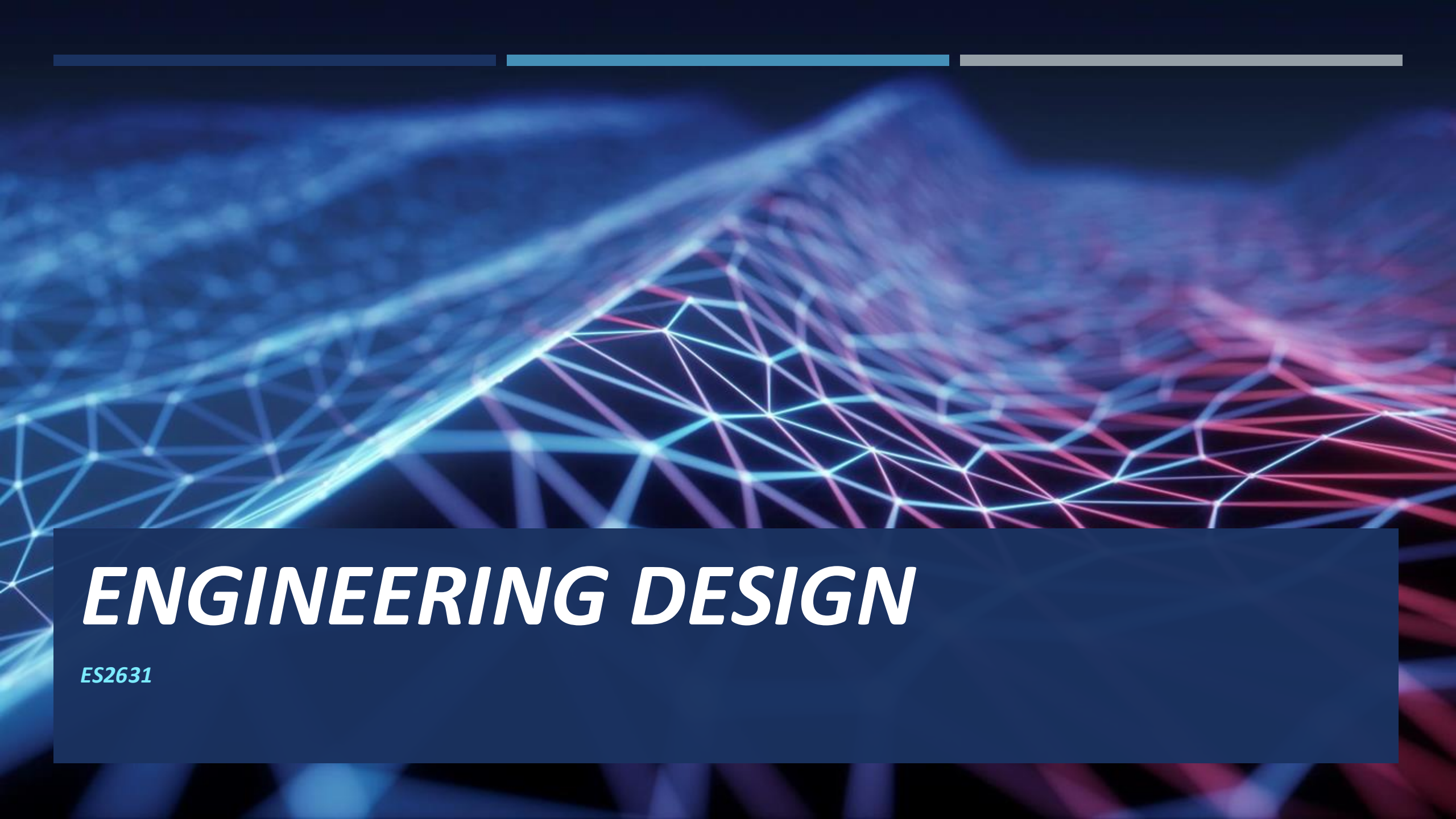
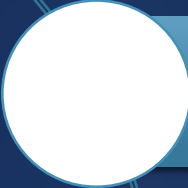




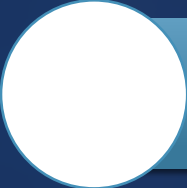
ENGINEERING DESIGN

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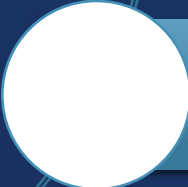
AGENDA



Basic Engineering Design



Project on Engineering Design



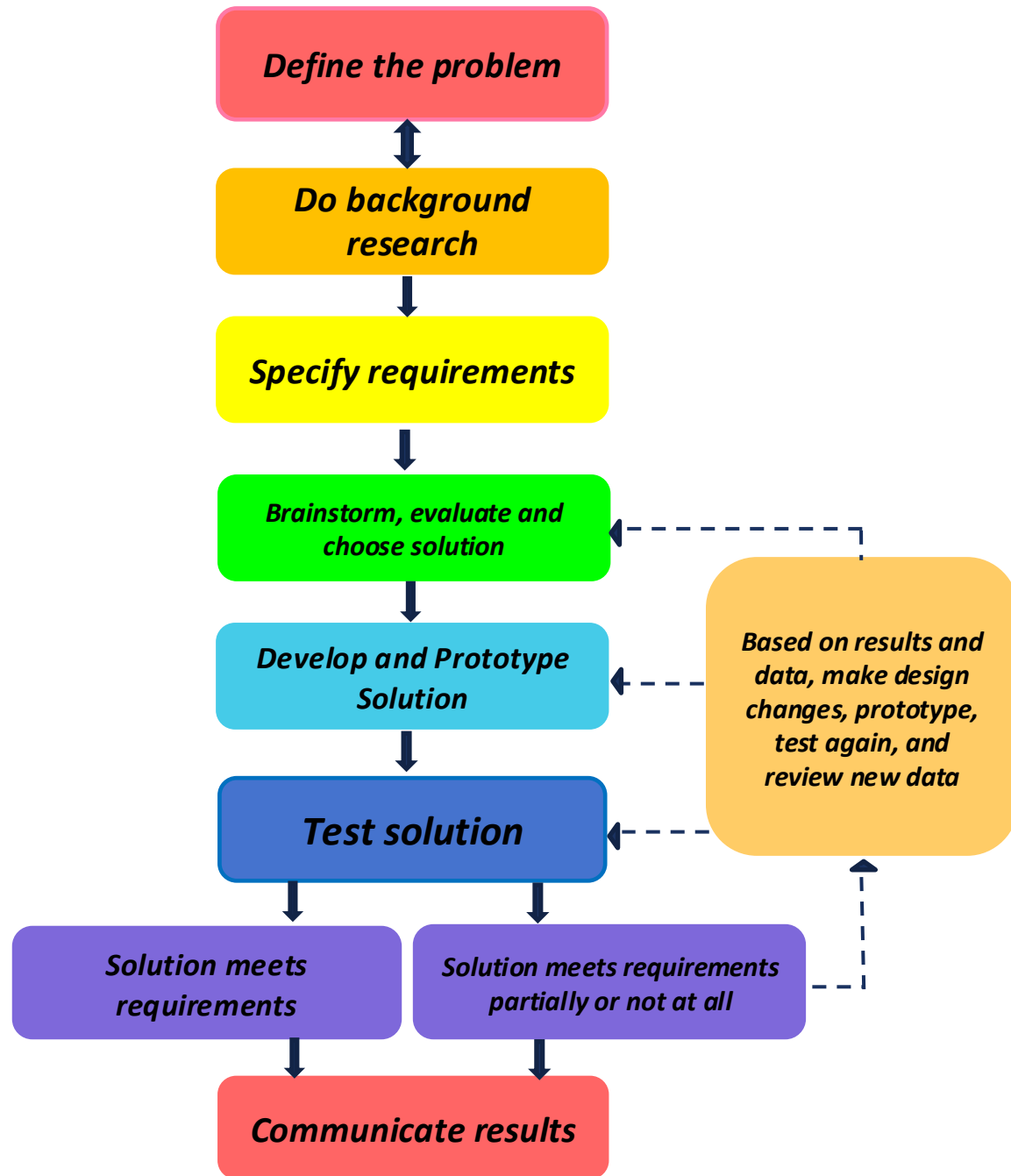
Ethics in Engineering Design



BASIC ENGINEERING DESIGN

PROCESS





1. Define the problem

- What is the problem/need?
- Who has the problem/need?
- Why is it important to solve?

2. Do background research

- Who are the main users?
- Are there existing solutions to similar problems?
[Avoid mistakes made]

3. Specify requirements

- What are important characteristics the solution must have?

4. Brainstorm solutions

- What are all the possible solutions?

5. Develop the solution

6. Build a prototype

7. Test and redesign

1. ASK to identify the need & constraints.

- *What is the problem to solve?*
- *What do we want to design?*
- *Who is it for?*
- *What do we want to accomplish?*

2. RESEARCH the problem.

- *What products or solutions already exist?*
- *What technologies might be adaptable to the identified need?*

3. IMAGINE/THINK of possible solutions.

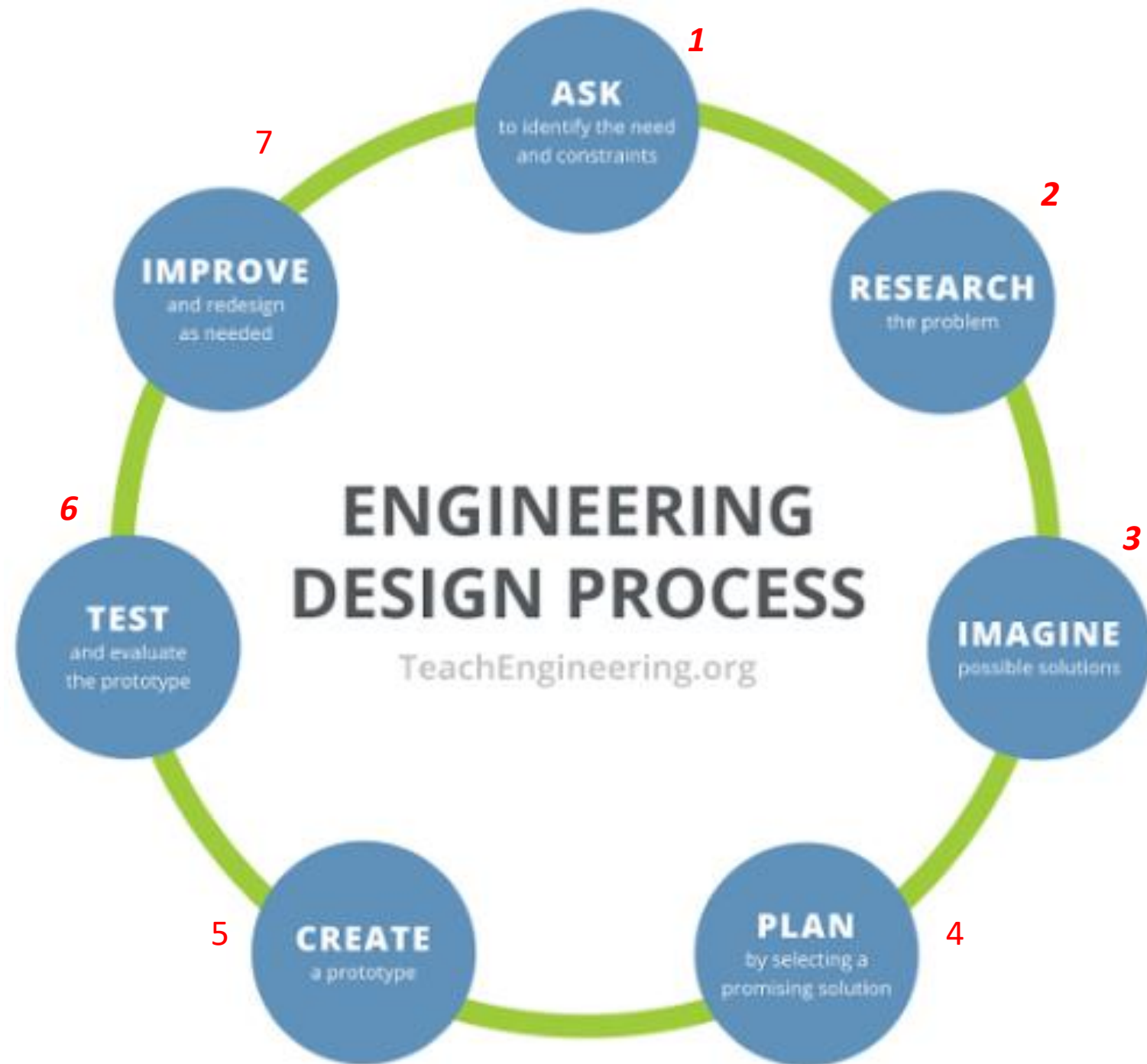
- *Brainstorm ideas: come up with as many solutions as possible*
- *Compare the solutions with the needs, constraints and research above*

4. PLAN by selecting a promising solution.

5. CREATE a prototype.

6. TEST and evaluate the prototype.

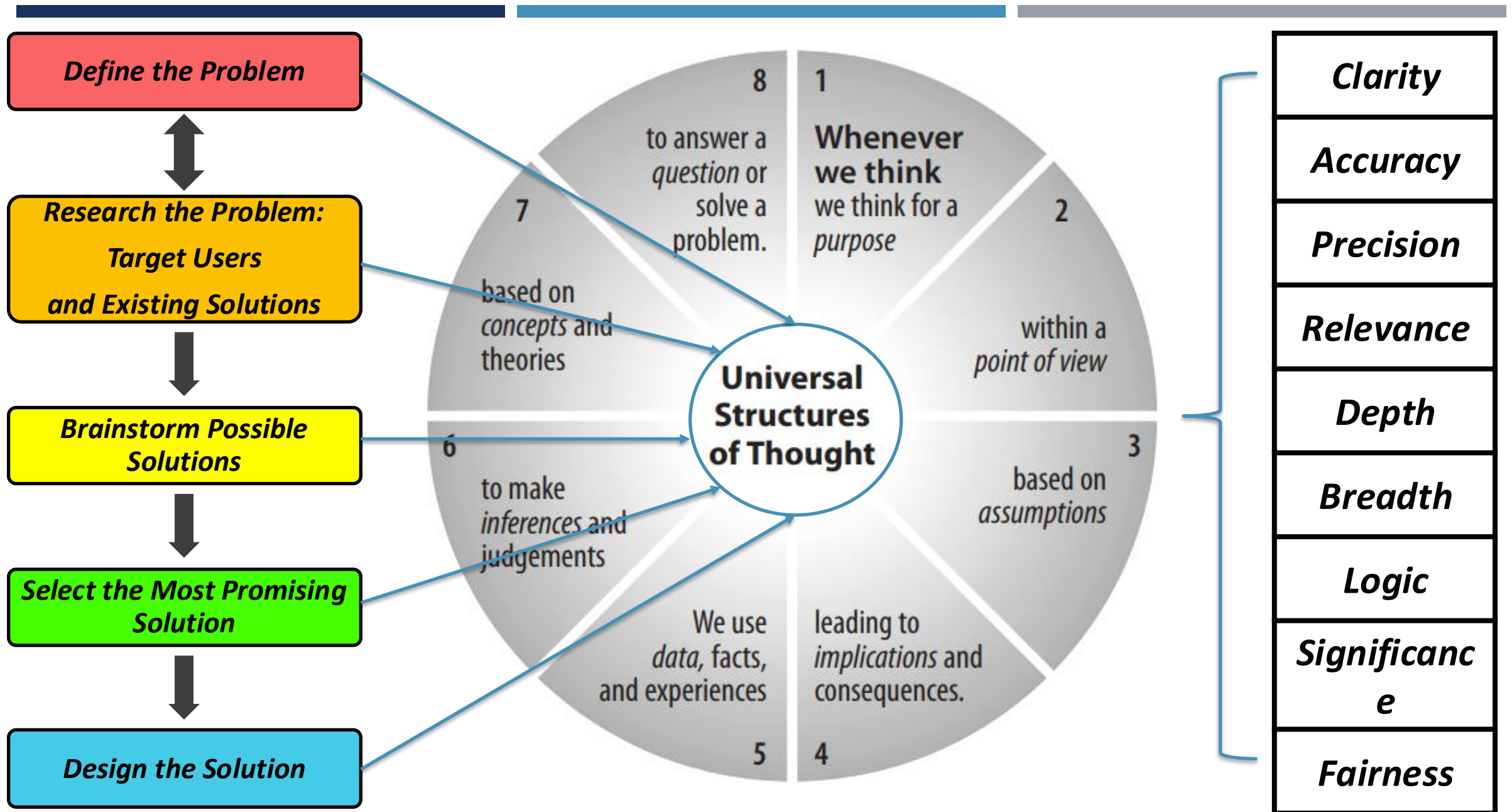
7. IMPROVE and redesign as needed.

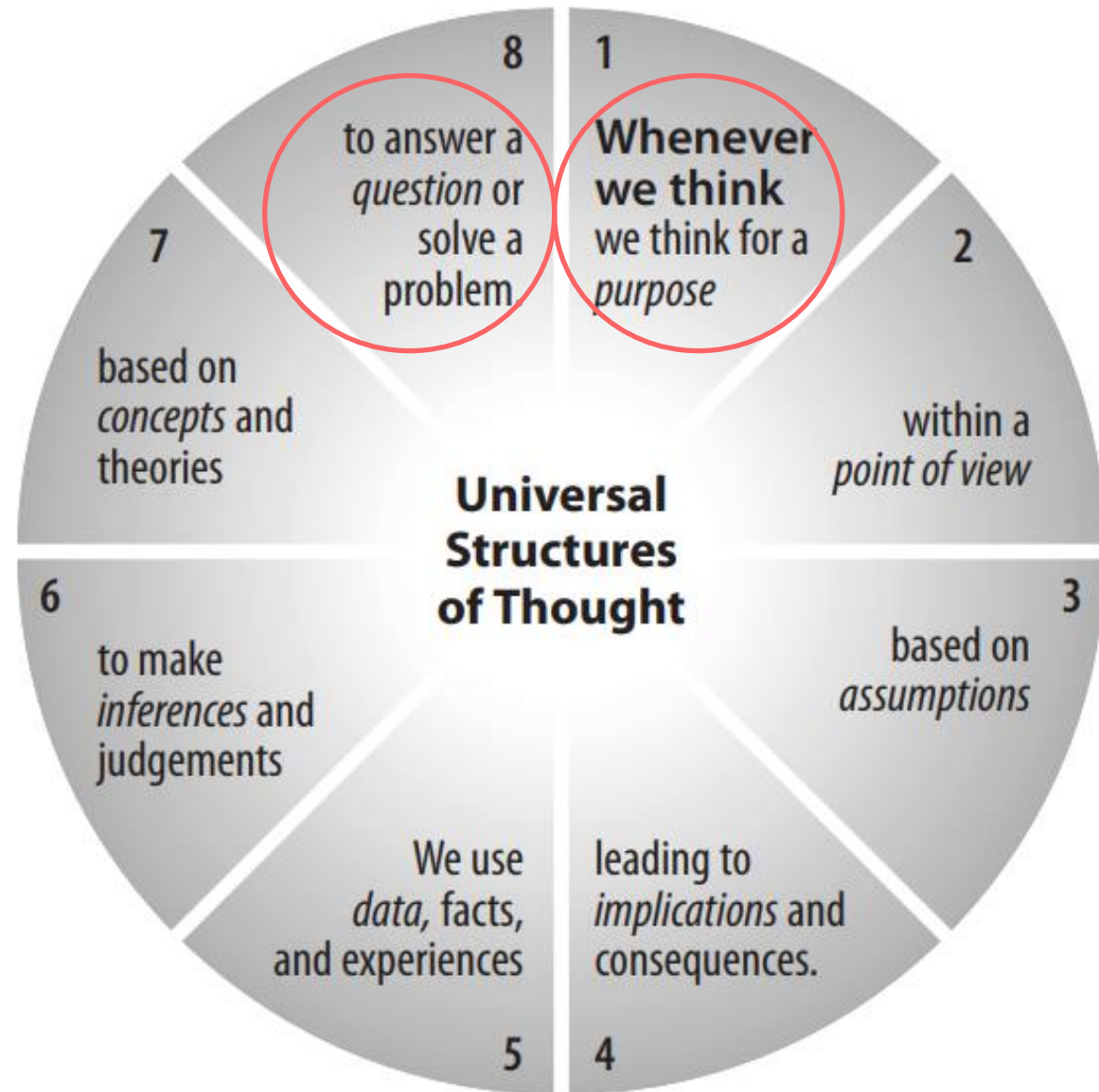
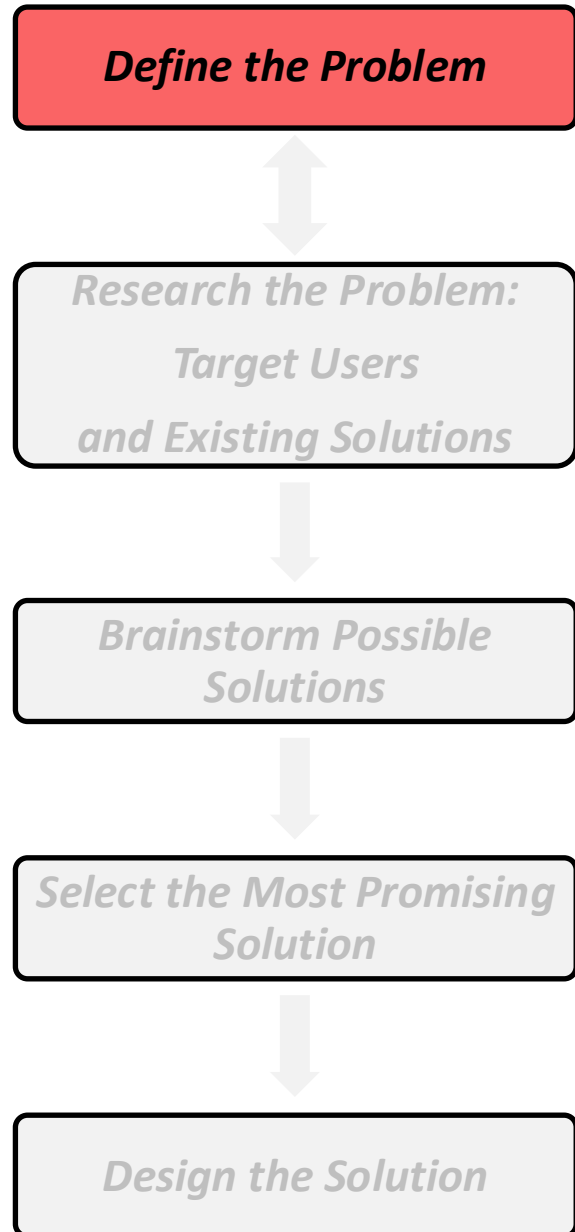


CONCEPTUAL PROJECT ON ENGINEERING DESIGN



ASSIGNMENT 1





QUESTIONS TO CONSIDER WHEN DEFINING THE PROBLEM

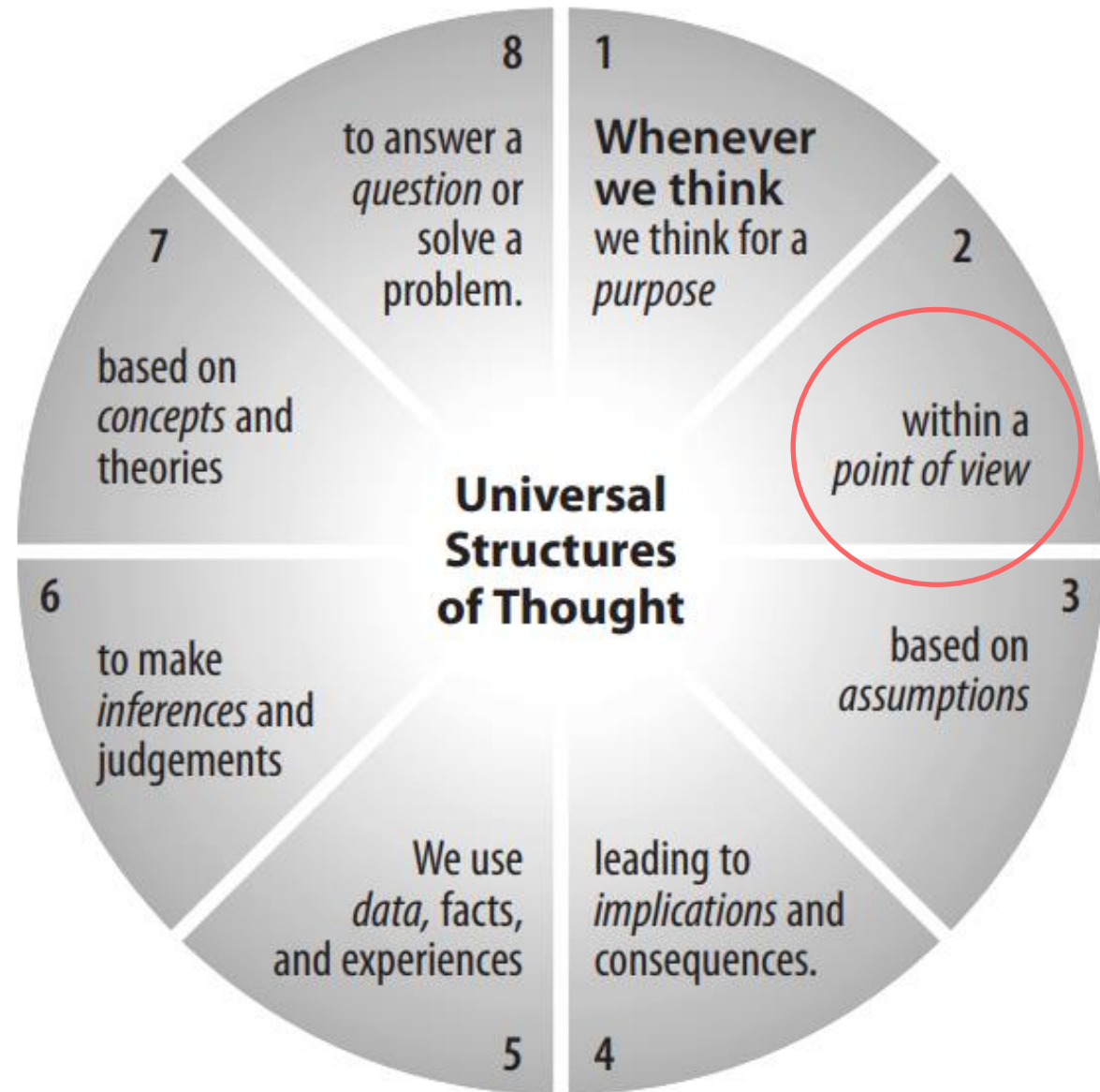
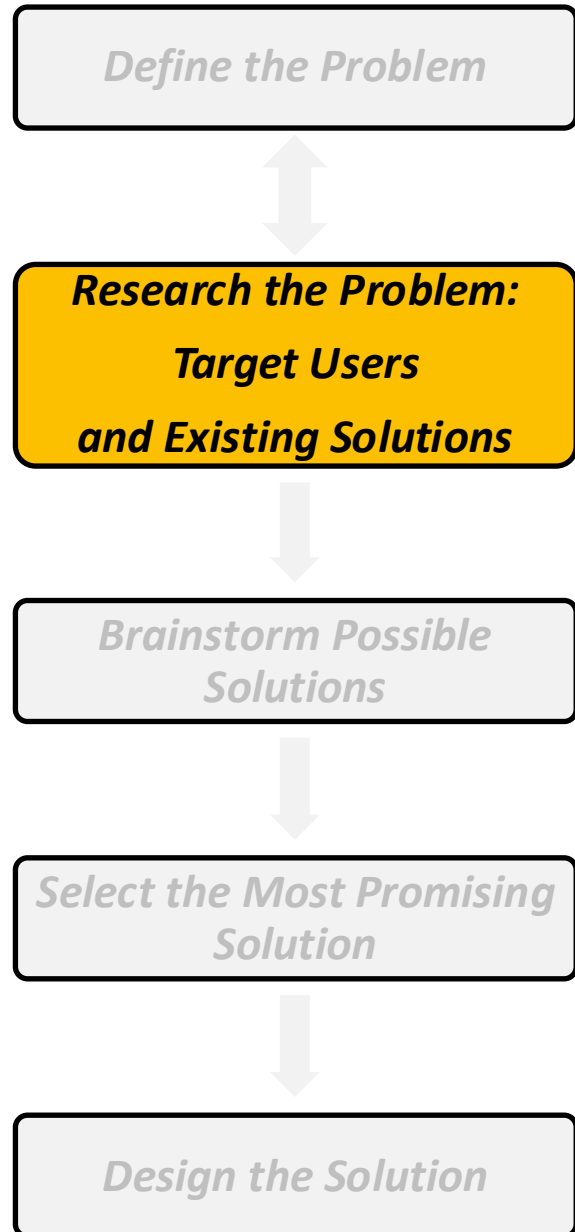
Question/Problem

- *Have I stated the question at issue clearly and precisely?*
- *Have I expressed the question in several ways to clarify its meaning and scope?*
- *Have I divided the question into sub-questions?*
 - *What system/product/process will best satisfy the user's performance, cost, and schedule requirements?*
- *Have I determined if the question has one right answer, or requires reasoning from more than one hypothesis or point of view?*
 - *Is a new design or new technology required? Can an existing design be adapted?*

QUESTIONS TO CONSIDER WHEN DEFINING THE PROBLEM

Purpose

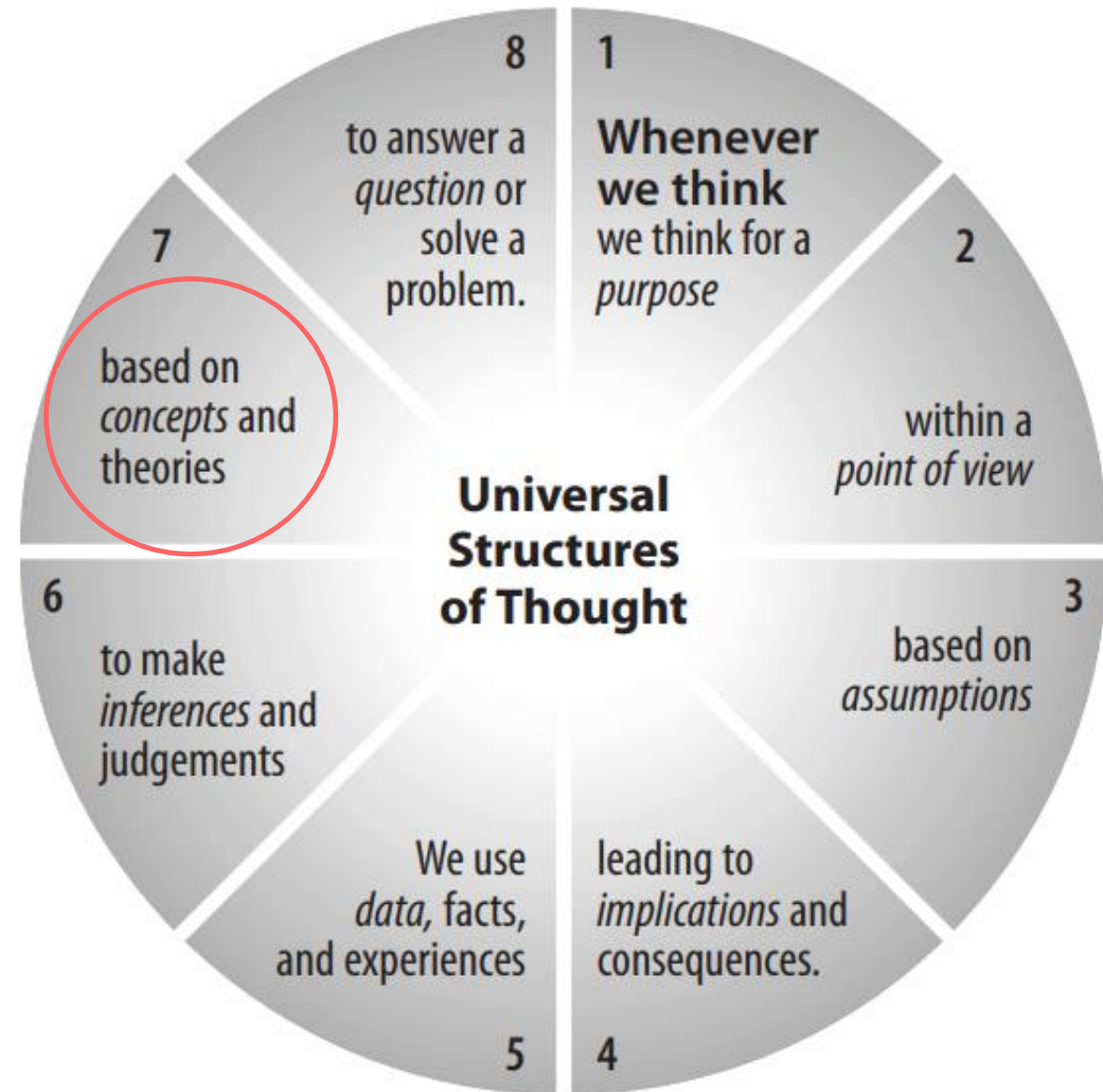
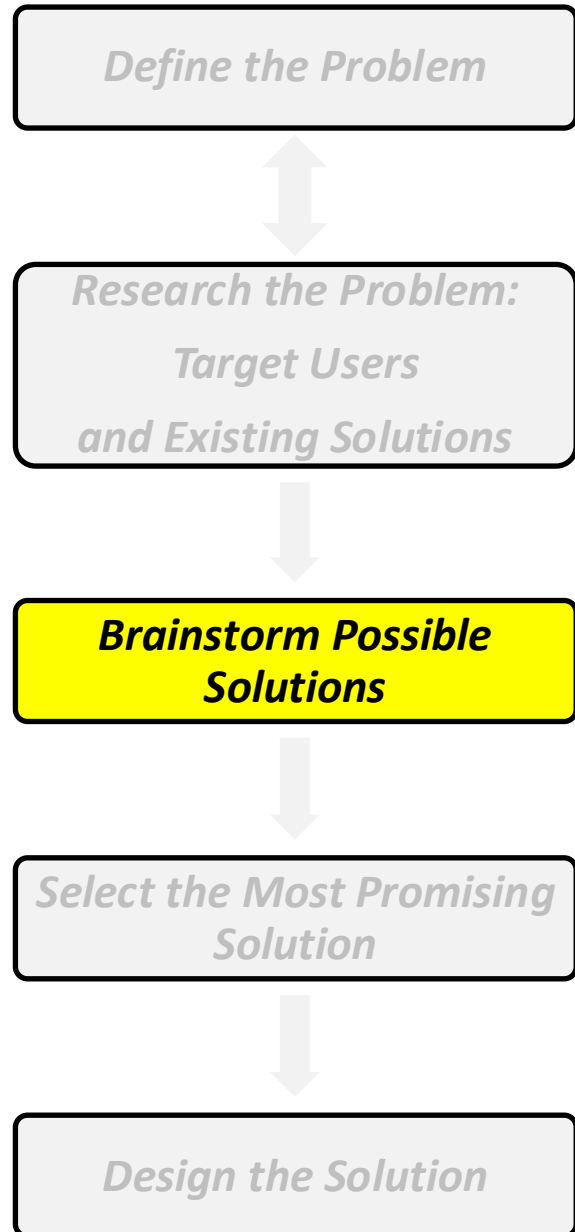
- *Have I distinguished my purpose from related purposes?*
 - *What are the mission requirements? Who defines them?*
 - *Who is the target audience? Who are the users?*
 - *What is the purpose of this design?*
- *Have I checked periodically to be sure I am still on target?*
- *Have I chosen realistic and achievable purposes?*



QUESTIONS TO CONSIDER WHEN CONSIDERING TARGET USERS

Point of View

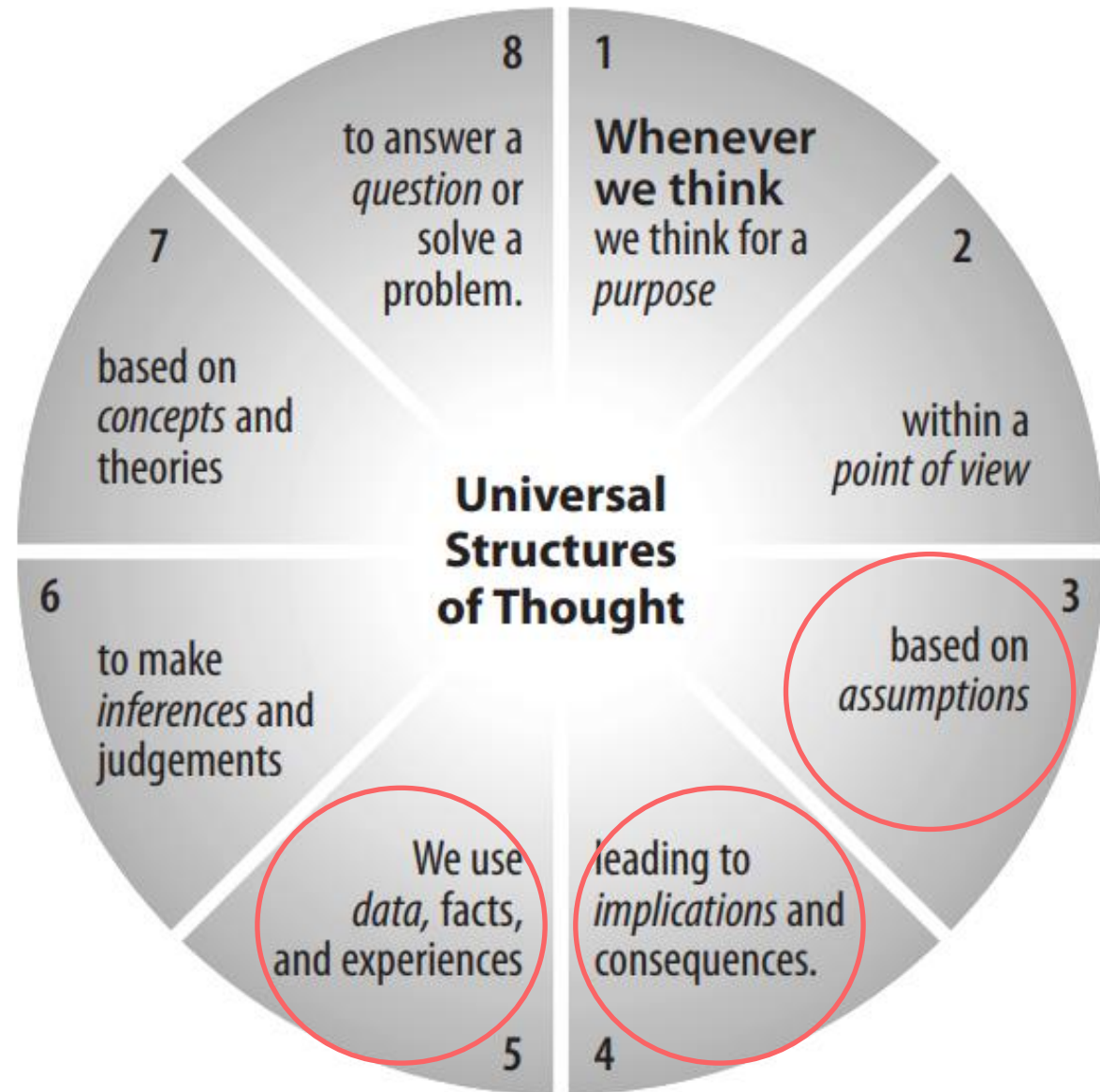
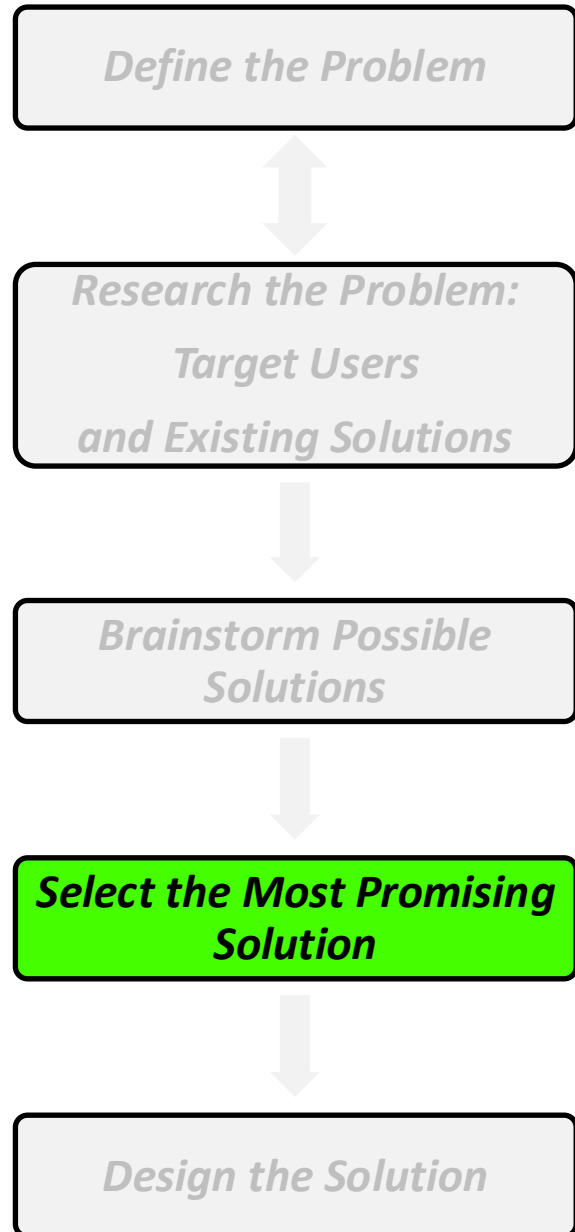
- *Have I considered the point of view of other stakeholders?*
- *Have I striven to be fairminded in evaluating all relevant points of view?*



QUESTIONS TO CONSIDER WHEN BRAINSTORMING POSSIBLE SOLUTIONS

Concepts

- *Have I identified key concepts and explained them clearly?*
 - *What concepts or theories are applicable to this problem?*
 - *What available technologies or theories are appropriate?*
- *Have I considered alternative concepts or alternative definitions of concepts?*
 - *Are there competing models?*



QUESTIONS TO CONSIDER WHEN SELECTING THE MOST PROMISING SOLUTION

Data and Information

- *Have I ensured that I have gathered sufficient data?*
 - *Is the available information sufficient? Do we need more data? What is the best way to collect it? Survey analysis? Simulation?*
- *Have I searched for data that opposes my position as well as alternative theories?*
 - *What legacy solutions, shortcomings, or problems should be studied and evaluated?*

QUESTIONS TO CONSIDER WHEN SELECTING THE MOST PROMISING SOLUTION

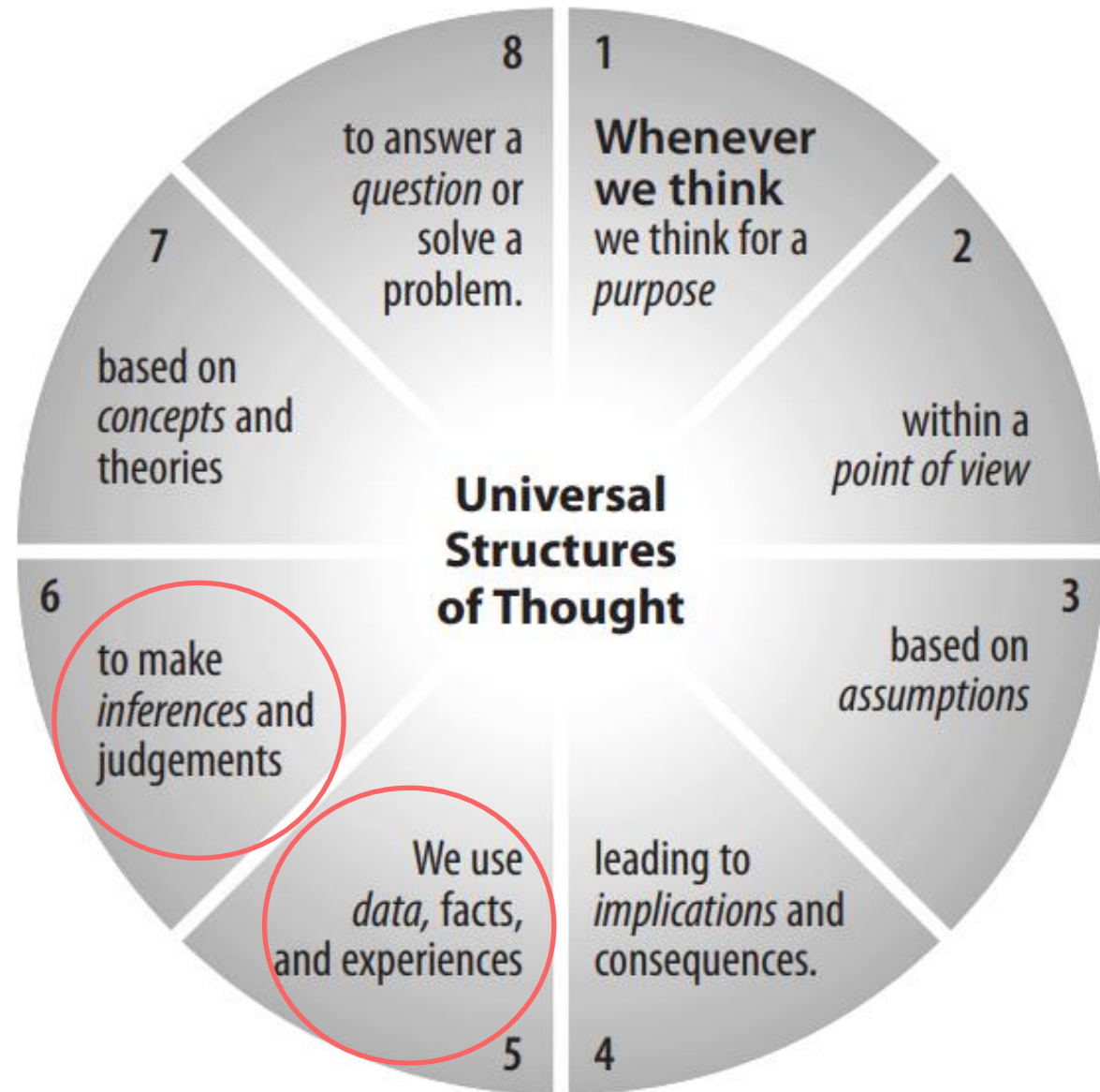
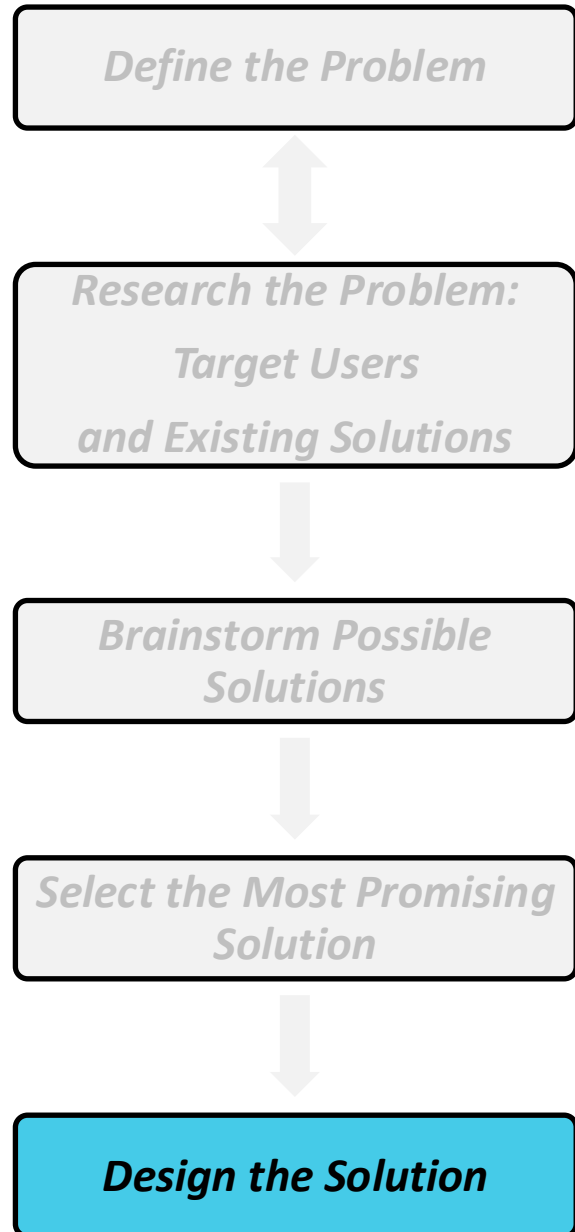
Assumptions

- *Have I clearly identified my assumptions and determined whether they are justifiable?*
 - *What market/economic/competitive environment is assumed? What safety/environmental assumptions are we making?*
 - *What criteria have historically been assumed in defining a “best” or “optimum” solution?*
- *Have I considered how my assumptions are shaping my point of view?*
- *Have I considered which of my assumptions might be reasonably questioned?*
 - *What programmatic, financial, market or technical risks have been considered acceptable to date?*
- *Are these assumptions acceptable? What happens if we change or discard an assumption?*

QUESTIONS TO CONSIDER WHEN SELECTING THE MOST PROMISING SOLUTION

Implications

- *What are some important implications of the data we have gathered?*
 - *Have I considered all significant implications?*
- *Have I traced the implications that follow from the data and from my reasoning?*
- *Have I searched for negative as well as positive implications (technical, social, environmental, financial, ethical)?*
 - *What are the most important implications of product failure?*
 - *Should social reaction and change management issues be addressed?*



QUESTIONS TO CONSIDER WHEN DESIGNING THE SOLUTION

Data and Information

- *Have I validated my data sources?*
 - *What is the source of supporting information? Have we considered all relevant sources?*
- *Have I restricted my claims to those supported by the data?*
- *Have I ensured that all data used is clear, accurate, and relevant to the question at issue?*

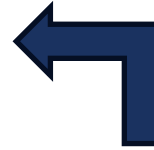
QUESTIONS TO CONSIDER WHEN DESIGNING THE SOLUTION

Inferences

- *Have I inferred only what the data supports?*
 - *What is the set of viable candidate solutions?*
 - *Is the conclusion practicable and affordable?*
- *Have I checked inferences for their internal and external consistency?*
- *Have I identified assumptions that led to my conclusions?*
- *Is there another way to interpret the information?*
 - *Why were other candidate solutions rejected?*

A Checklist for Engineering Reasoning

1. All engineering reasoning expresses a *purpose*.
 - Have I distinguished my purpose from related purposes?
 - Have I checked periodically to be sure I am still on target?
 - Have I chosen realistic and achievable purposes?
2. All engineering reasoning seeks to figure something out, to settle some *question*, solve some *engineering problem*.
 - Have I stated the question at issue clearly and precisely?
 - Have I expressed the question in several ways to clarify its meaning and scope?
 - Have I divided the question into sub-questions?
 - Have I determined if the question has one right answer, or requires reasoning from more than one hypothesis or point of view?
3. All engineering reasoning requires *assumptions*.
 - Have I clearly identified my assumptions and determined whether they are justifiable?
 - Have I considered how my assumptions are shaping my point of view?
 - Have I considered which of my assumptions might be reasonably questioned?
4. All engineering reasoning is done from some *perspective or point of view*.
 - Have I identified my specific point of view?
 - Have I considered the point of view of other stakeholders?
 - Have I striven to be fairminded in evaluating all relevant points of view?
5. All engineering reasoning is based on *data, information, and evidence*.
 - Have I validated my data sources?
 - Have I restricted my claims to those supported by the data?
 - Have I searched for data that opposes my position as well as alternative theories?
 - Have I ensured that all data used is clear, accurate, and relevant to the question at issue?
 - Have I ensured that I have gathered sufficient data?
6. All engineering reasoning is expressed through, and shaped by, *concepts and theories*.
 - Have I identified key concepts and explained them clearly?
 - Have I considered alternative concepts or alternative definitions of concepts?
 - Have I distorted ideas to fit my agenda?
7. All engineering reasoning entails *inferences or interpretations* by which we draw *conclusions* and give meaning to engineering data and work.
 - Have I inferred only what the data supports?
 - Have I checked inferences for their internal and external consistency?
 - Have I identified assumptions that led to my conclusions?
8. All engineering reasoning leads somewhere or has *implications and consequences*.
 - Have I traced the implications that follow from the data and from my reasoning?
 - Have I searched for negative as well as positive implications (technical, social, environmental, financial, ethical)?
 - Have I considered all significant implications?



CHECKLIST FOR ENGINEERING REASONING

&

ANALYSING A DESIGN USING THE ELEMENTS OF THOUGHT



Analyzing a Design Using the Elements of Thought

Engineering purpose	What is the purpose of this design? What are the market opportunities or mission requirements? Who defines market opportunities/mission requirements? Who is the customer?
Question at hand	What system/product/process will best satisfy the customer's performance, cost, and schedule requirements? How does the customer define "value"? Is a new design or new technology required? Can an existing design be adapted? How important is time-to-market?
Point of view	A design and manufacturing point of view is typically presumed. What other points of view deserve consideration? Stockholders? Component vendors/suppliers? Marketing/sales? Customers? Maintenance/repair/parts? Regulators? Community affairs? Politicians? Environmentalists?
Assumptions	What environmental or operating conditions are assumed? What programmatic, financial, market or technical risks have been considered acceptable to date? What market/economic/competitive environment is assumed? What safety/environmental assumptions are we making? Are these assumptions acceptable? What maturity level or maturation timeline is assumed for emerging technologies? What happens if we change or discard an assumption? What criteria have historically been assumed in defining a "best" or "optimum" solution? What assumptions have been made on the availability of materials? What manufacturing capability was assumed? What workforce skills or attributes have been assumed?

ETHICS IN ENGINEERING DESIGN

Ethical implications of normal operations, possible failure modes and misuse by user.

- 1. Humanitarian responsibility and product safety***
- 2. Fiduciary responsibility***
- 3. Environmental responsibility***

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